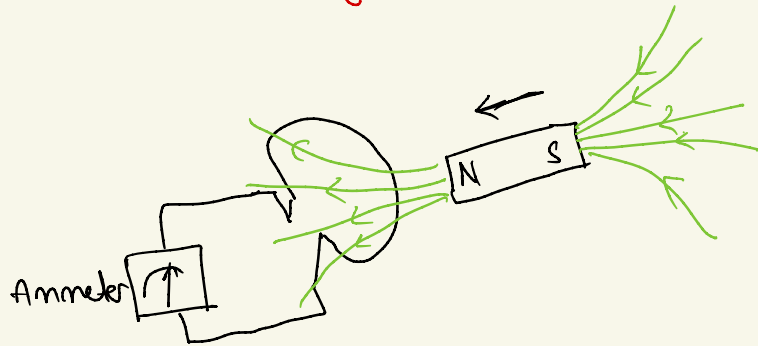


# Resnick Ch 30: Induction and Inductance

Michael Faraday 1791 - 1867

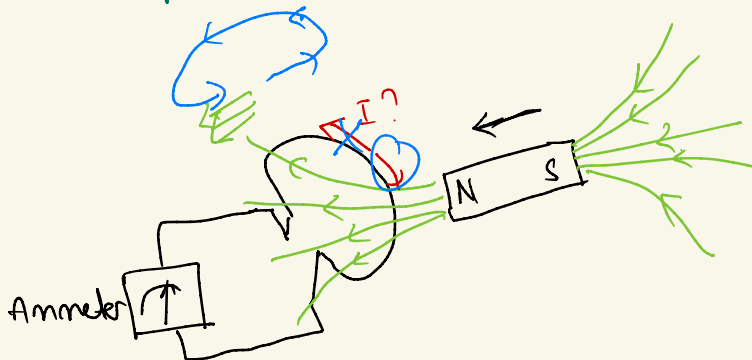


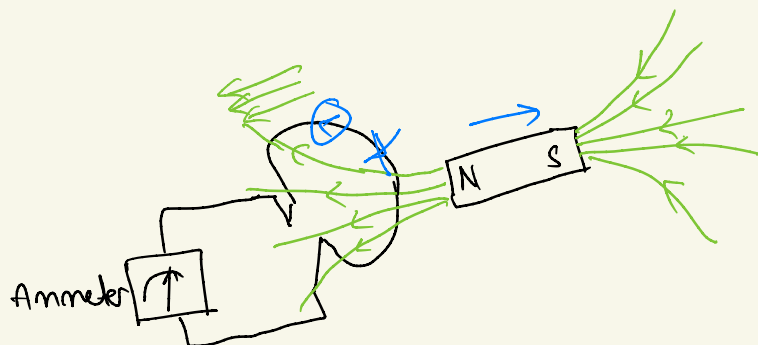
$$-\frac{d\Phi_B}{dt} = \text{EMF (Voltage)}$$

$$\Phi_B = \text{magnetic flux} = BA$$

Heinrich Lenz 1804 - 1865

Result: EMF is induced in such a way that it tries to keep initial stage.





Multiple wires :

$$\mathcal{E} = -N \frac{d\Phi_B}{dt}$$

**27 ILW** As seen in Fig. 30-48, a square loop of wire has sides of length 2.0 cm. A magnetic field is directed out of the page; its magnitude is given by  $B = 4.0t^2y$ , where  $B$  is in teslas,  $t$  is in seconds, and  $y$  is in meters. At  $t = 2.5$  s, what are the (a) magnitude and (b) direction of the emf induced in the loop?

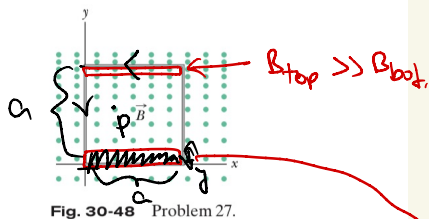


Fig. 30-48 Problem 27.

$$\mathcal{E} = - \frac{d\Phi_B}{dt}$$

$$\Phi_B = B \cdot A$$

$$= (4.0t^2y) \text{ Area}$$

$$\Phi_B = \int \Phi_B$$

$$d\Phi_B = \int B \cdot dA$$

$$= \int (4.0t^2y) a \, dy$$

$$= 4.0t^2 a \int_0^a y \, dy$$

$$= 4.0t^2 a \cdot \frac{a^2}{2}$$

$$\Phi_B(t) = 2.0t^2 a^3$$

a)

$$\mathcal{EMF} = - \frac{d\Phi_B}{dt}$$

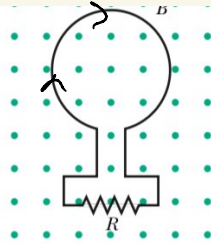
$$= - \frac{d}{dt} (2t^2 a^3) = - (4ta^3) \rightarrow \mathcal{E}(t = 2.5\text{s}) = \frac{-4 \cdot 2.5 \cdot (0.02\text{m})^3}{8 \times 10^{-5}\text{V}}$$

b) counter clockwise : to compensate increasing B-field (flux)

$$B(t) = 4.0t^2 \hat{y}$$

↑  
increasing with time!

•7 In Fig. 30-36, the magnetic flux through the loop increases according to the relation  $\Phi_B = 6.0t^2 + 7.0t$ , where  $\Phi_B$  is in milliwbebers and  $t$  is in seconds. (a) What is the magnitude of the emf induced in the loop when  $t = 2.0$  s? (b) Is the direction of the current through  $R$  to the right or left?



out of the page

Fig. 30-36 Problem 7.

•8 A uniform magnetic field  $\vec{B}$  is perpendicular to the plane of a circular

$$\text{Area} = \text{const.}$$

$$\Phi_B = 6.0t^2 + 7.0t$$

$$(a) \quad \mathcal{E} = - \frac{d\Phi_B}{dt} = -3V$$

(b) clockwise

•18 In Fig. 30-43, two straight conducting rails form a right angle. A conducting bar in contact with the rails starts at the vertex at time  $t = 0$  and moves with a constant velocity of 5.20 m/s along them. A magnetic field with  $B = 0.350$  T is directed out of the page. Calculate (a) the flux through the triangle formed by the rails and bar at  $t = 3.00$  s and (b) the emf around the triangle at that time. (c) If the emf is  $\mathcal{E} = at^n$ , where  $a$  and  $n$  are constants, what is the value of  $n$ ?

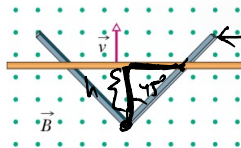
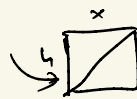


Fig. 30-43 Problem 18.

$$v = \text{const.}$$

conducting rails

$$(a) \quad \Phi = BA$$



$$h = vt$$

$$x = h = vt$$

$$\text{Area} = \left( \frac{h \cdot x}{2} \right) = \frac{(vt)^2}{2}$$

$$\Phi_B = B \cdot A = Bvt^2$$

$$= (0.35T)(5.20 \frac{m}{s})^2 (3s)$$

$$= 2$$

$$b) \quad \mathcal{E} = - \frac{d\Phi_B}{dt}$$

$$= - \frac{d}{dt}(Bvt^2) = -(2Bvt)$$

$$c) \quad n = 1$$

$$\mathcal{E} = at^n = 2Bvt$$

$$n = 1$$

••26 For the wire arrangement in Fig. 30-47,  $a = 12.0$  cm and  $b = 16.0$  cm. The current in the long straight wire is  $i = 4.50t^2 - 10.0t$ , where  $i$  is in amperes and  $t$  is in seconds. (a) Find the emf in the square loop at  $t = 3.00$  s. (b) What is the direction of the induced current in the loop?

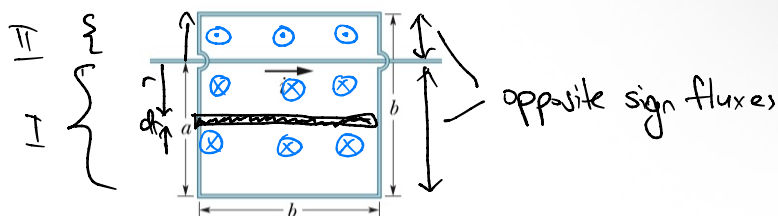


Fig. 30-47 Problem 26.

$$\Phi_B = \int d\Phi_B$$

$$= \underbrace{\int_0^a \frac{\mu_0 i b}{2\pi} \frac{dr}{r}}_I - \underbrace{\int_0^{b-a} \frac{\mu_0 i b}{2\pi} \frac{dr}{r}}_{II}$$

$$= \frac{\mu_0 i b}{2\pi} \ln r \Big|_0^a - \frac{\mu_0 i b}{2\pi} \ln r \Big|_0^{b-a}$$

$$= \frac{\mu_0 i b}{2\pi} \ln \left( \frac{a}{b-a} \right)$$

$$\begin{aligned} d\Phi_B &= B(r) (b dr) \\ &= \frac{\mu_0 i}{2\pi r} (b dr) \\ &= \frac{\mu_0 i b}{2\pi r} dr \end{aligned}$$

$$\begin{aligned} &\ln(a) - \ln(b) - (\ln(c) - \ln(d)) \\ &\ln a - \ln b - \ln(b-a) + \ln b \\ &\ln a - \ln(b-a) \\ &\ln \left( \frac{a}{b-a} \right) \end{aligned}$$

no  
doesn't  
exist  
actually

$$\mathcal{E} = - \frac{d\Phi_B}{dt}$$

$$= - \frac{d}{dt} \left( \frac{\mu_0 i b}{2\pi} \ln \left( \frac{a}{b-a} \right) \right)$$

$$= \frac{\mu_0 b}{2\pi} \ln \left( \frac{a}{b-a} \right) \left[ \frac{di}{dt} \right]$$

$$i = 4.5t^2 - 10t$$

$$\frac{di}{dt} = 9t - 10$$

# Official Solution:

26. (a) First, we observe that a large portion of the figure contributes flux that “cancels out.” The field (due to the current in the long straight wire) through the part of the rectangle above the wire is out of the page (by the right-hand rule) and below the wire it is into the page. Thus, since the height of the part above the wire is  $b - a$ , then a strip below the wire (where the strip borders the long wire, and extends a distance  $b - a$  away from it) has exactly the equal but opposite flux that cancels the contribution from the part above the wire. Thus, we obtain the non zero contributions to the flux:

$$\Phi_B = \int B dA = \int_{b-a}^a \left( \frac{\mu_0 i}{2\pi r} \right) (b dr) = \frac{\mu_0 i b}{2\pi} \ln \left( \frac{a}{b-a} \right).$$

Faraday’s law, then, (with SI units and 3 significant figures understood) leads to

$$\begin{aligned} \varepsilon &= -\frac{d\Phi_B}{dt} = -\frac{d}{dt} \left[ \frac{\mu_0 i b}{2\pi} \ln \left( \frac{a}{b-a} \right) \right] = -\frac{\mu_0 b}{2\pi} \ln \left( \frac{a}{b-a} \right) \frac{di}{dt} \\ &= -\frac{\mu_0 b}{2\pi} \ln \left( \frac{a}{b-a} \right) \frac{d}{dt} \left( \frac{9}{2} t^2 - 10t \right) \\ &= \frac{-\mu_0 b (9t - 10)}{2\pi} \ln \left( \frac{a}{b-a} \right). \end{aligned}$$

With  $a = 0.120$  m and  $b = 0.160$  m, then, at  $t = 3.00$  s, the magnitude of the emf induced in the rectangular loop is

$$|\varepsilon| = \frac{(4\pi \times 10^{-7})(0.16)(9(3) - 10)}{2\pi} \ln \left( \frac{0.12}{0.16 - 0.12} \right) = 5.98 \times 10^{-7} \text{ V}.$$

(b) We note that  $di/dt > 0$  at  $t = 3$  s. The situation is roughly analogous to that shown in Fig. 30-5(c). From Lenz’s law, then, the induced emf (hence, the induced current) in the loop is counterclockwise.

Resistor is a dissipative element



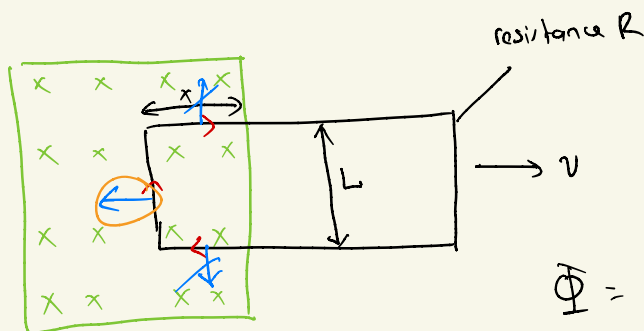
$$\mathcal{E} = iR \rightarrow i = \frac{\mathcal{E}}{R}$$

$$(dq) V$$

$$(dq) iR = dW$$

$$\frac{dW}{dt} = P = \left(\frac{dq}{dt}\right) iR = i^2 R$$

$$W = i^2 R t = V i t$$



$\Phi$  is decreasing

$B = \text{const}$

$A = \text{decreasing}$

$$\Phi = B \cdot Lx$$

$$-\frac{d\Phi}{dt} = -BL \frac{dx}{dt} = -BLv$$

$$\vec{F}_B = i \vec{l} \times \vec{B}$$

$$\mathcal{E} = -BLv = iR$$

$$i = -\frac{BLv}{R}$$

$$\text{Work} = \vec{F} \cdot d\vec{r}$$

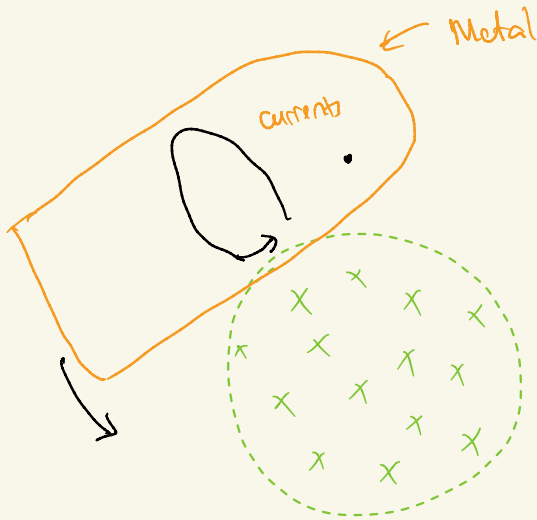
$$= \vec{F}_B \cdot d\vec{x} = iLB(\sin 90^\circ) dx$$

$$= \int_0^x \left(\frac{BLv}{R}\right) LB dx$$

## Eddy currents

So far, we have checked wires.

How about pieces of metal plates?



Overall changing B-field  $\Rightarrow$  E-field (EMF)

Индуктивность (капакан)

$$L = \frac{N\Phi_B}{i}$$

(definition of inductance)

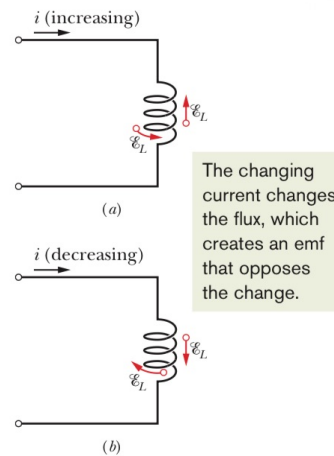
$$\mathcal{E} = - \frac{d\Phi_B}{dt} \quad (= - N \frac{d\Phi_B}{dt} \text{ when many turns of coils})$$

$$Li = N\Phi_B$$

$$\frac{d(Li)}{dt} = \frac{d(N\Phi_B)}{dt} = N \frac{d\Phi_B}{dt} = -\mathcal{E} \rightarrow \boxed{\mathcal{E} = -L \frac{di}{dt}}$$

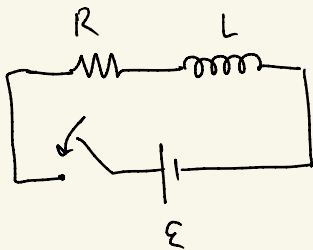
Same logic: Lenz's Law

- Tries to recover (maintain) initial state

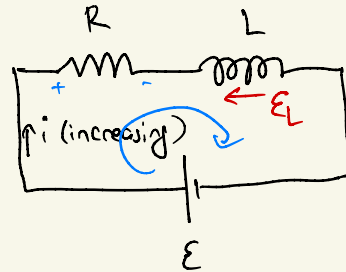


**Fig. 30-14** (a) The current  $i$  is increasing, and the self-induced emf  $\mathcal{E}_L$  appears along the coil in a direction such that it opposes the increase. The arrow representing  $\mathcal{E}_L$  can be drawn along a turn of the coil or alongside the coil. Both are shown. (b) The current  $i$  is decreasing, and the self-induced emf appears in a direction such that it opposes the decrease.

RL - Circuits



$\Rightarrow$



$$+\mathcal{E} - iR - L \frac{di}{dt} = 0$$

$$(\mathcal{E} - iR) - L \frac{di}{dt} = 0$$

$$\frac{(\mathcal{E} - iR)}{L} dt = \frac{L di}{L}$$

$$\frac{di}{\mathcal{E} - iR} = \frac{dt}{L} \rightarrow$$

$$\frac{di}{\varepsilon - iR} = \frac{dt}{L} \rightarrow \int_0^i \frac{di}{i - \frac{\varepsilon}{R}} = - \int_0^t \frac{R}{L} dt$$

Use Method of substitution again: let  $u = i - \frac{\varepsilon}{R}$

$$\frac{du}{di} = 1.$$

$$\int \frac{du}{u} = - \frac{R}{L} t$$

$$\ln u \Big|_{u_i}^{u_f} = - \frac{R}{L} t$$

Put back  $u = i - \frac{\varepsilon}{R}$ :  $\ln \left( i - \frac{\varepsilon}{R} \right) \Big|_0^{I(t)} = - \frac{R}{L} t$

$$\ln \left( I - \frac{\varepsilon}{R} \right) - \ln \left( 0 - \frac{\varepsilon}{R} \right) = - \frac{R}{L} t$$

$$\ln \left( \frac{I - \frac{\varepsilon}{R}}{-\frac{\varepsilon}{R}} \right) = - \frac{R}{L} t$$

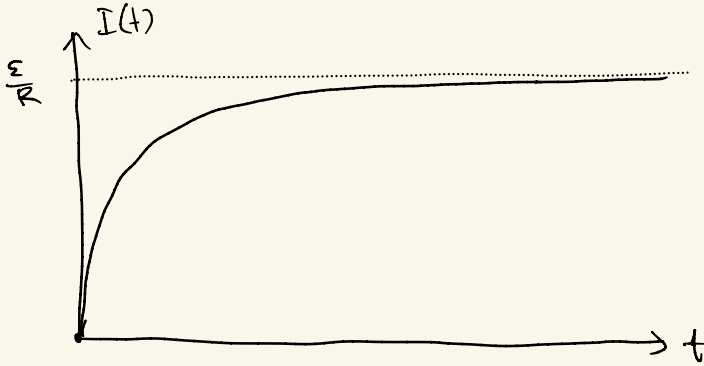
Take exp of both sides

$$\frac{I - \frac{\varepsilon}{R}}{-\frac{\varepsilon}{R}} = e^{-\frac{R}{L} t}$$

$$I - \frac{\varepsilon}{R} = \left( -\frac{\varepsilon}{R} \right) e^{-\frac{R}{L} t}$$

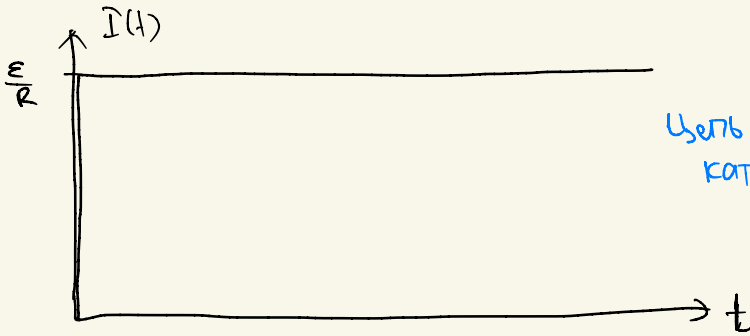
$$\boxed{I(t) = \frac{\varepsilon}{R} \left( 1 - e^{-t \frac{R}{L}} \right)}$$

$$I(t) = \frac{\varepsilon}{R} (1 - e^{-t \frac{R}{L}})$$



Цепь с  
катушкой

RL



Цепь без  
катушки

A bit about exponential functions:

$$I = I_0 e^{-\frac{t}{\tau}}$$

$\tau$ : time constant.



$\tau$ : time that  $I$  falls from

$$I_0 \rightarrow \frac{I_0}{e}$$

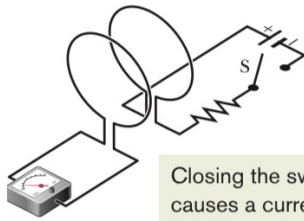
With the same logic

$$I(t) = \frac{\varepsilon}{R} (1 - e^{-t \frac{R}{L}})$$

in RL-circuits :

$$\tau = \frac{L}{R} \quad \text{time constant}$$

# Mutual Inductance



**Fig. 30-2** An ammeter registers a current in the left-hand wire loop just as switch S is closed (to turn on the current in the right-hand wire loop) or opened (to turn off the current in the right-hand loop). No motion of the coils is involved.

The mutual inductance of coil 2:

$$M_{21} = \frac{N_2 \Phi_{21}}{i_1}$$

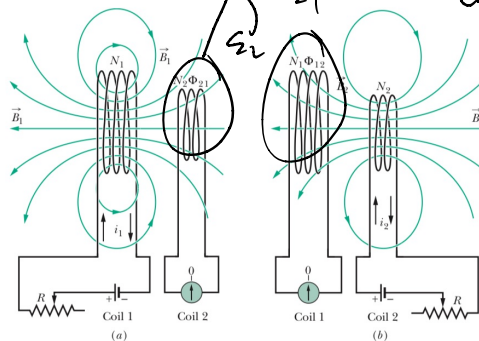
Rule:  $M_{21} = M_{12}$

Ex of Mutual Inductance: APHO 2011  
APHO 2025  
IPHO

We define the mutual inductance  $M_{21}$  of coil 2 with respect to coil 1 as

$$M_{21} = \frac{N_2 \Phi_{21}}{i_1}, \quad (30-57)$$

## 814 CHAPTER 30 INDUCTION AND INDUCTANCE



**Fig. 30-19** Mutual induction. (a) The magnetic field  $\vec{B}_1$  produced by current  $i_1$  in coil 1 extends through coil 2. If  $i_1$  is varied (by varying resistance  $R$ ), an emf is induced in coil 2 and current registers on the meter connected to coil 2. (b) The roles of the coils interchanged.

which has the same form as Eq. 30-28,

$$L = N\Phi/i, \quad (30-58)$$

the definition of inductance. We can recast Eq. 30-57 as

$$M_{21}i_1 = N_2\Phi_{21}. \quad (30-59)$$

If we cause  $i_1$  to vary with time by varying  $R$ , we have

$$M_{21} \frac{di_1}{dt} = N_2 \frac{d\Phi_{21}}{dt}. \quad (30-60)$$

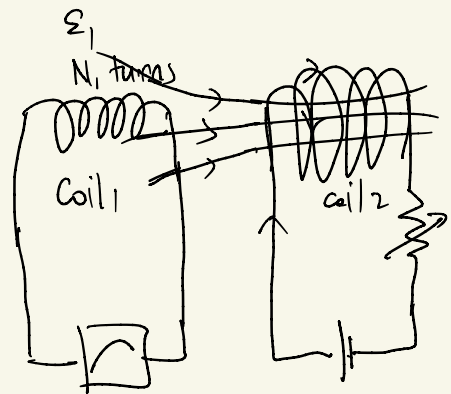
$$\mathcal{E}_1 = -M \frac{di_2}{dt}$$

$$\mathcal{E}_2 = -M \frac{di_1}{dt}$$

By definition:  $M_{21} = N_2 \frac{\Phi_{21}}{i_1}$  ,  $M_{12} = N_1 \frac{\Phi_{12}}{i_2}$

$$\mathcal{E}_2 = -\frac{d(N_2 \Phi_{21})}{dt} = -M_{21} \frac{di_1}{dt} = -M \frac{di_1}{dt}$$

Similarly  $\mathcal{E}_1 = -\frac{d(N_1 \Phi_{12})}{dt} = -M_{12} \frac{di_2}{dt}$



9) Figure 30-29 shows three circuits with identical batteries, inductors, and resistors. Rank the circuits, greatest first, according to the current through the resistor labeled  $R$  (a) long after the switch is closed, (b) just after the switch is reopened a long time later, and (c) long after it is reopened.

Fig. 30-28 Question 8.

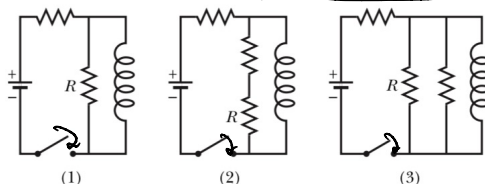
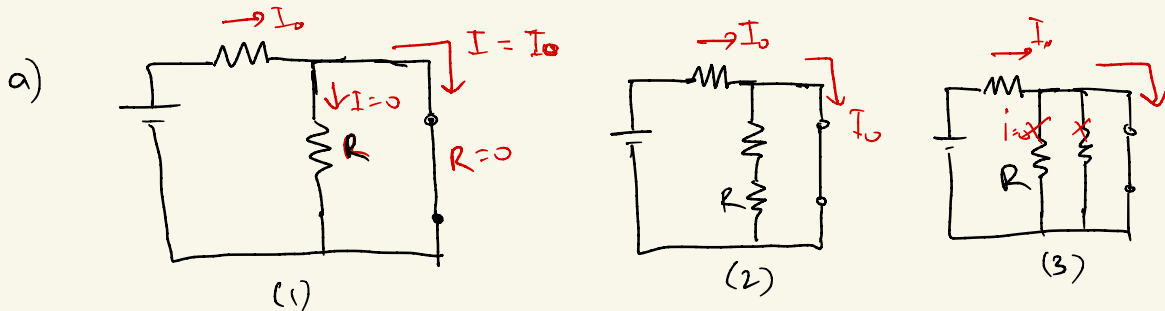
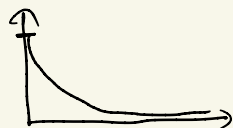
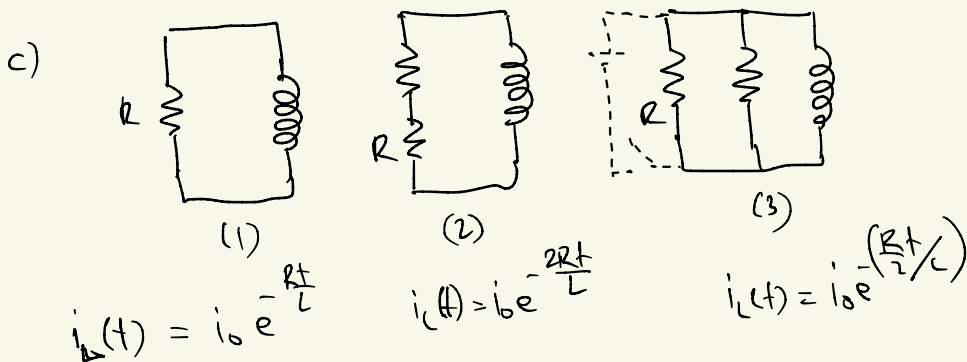


Fig. 30-29 Question 9.



$$i_{R(1)} = i_{R(2)} = i_{R(3)} = 0$$



$I_R = 0$  in all cases

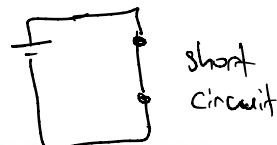
Thought for 1m 22s >

Idea: Treat the inductor as 

• a short circuit after a long time with the switch closed (dc steady state),

• a current source that keeps its current unchanged at the instant the switch is opened, and

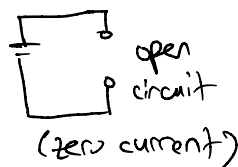
• an open circuit long after the switch is opened (current has decayed to zero).



short circuit

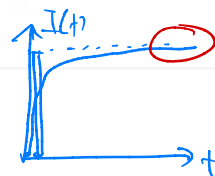
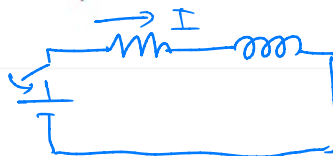
Also note the wiring differences:

- (1) has a hard short in parallel with  $R$  (even without the inductor).
- (2) has the inductor in parallel with a single  $R$ .
- (3) has the inductor in parallel with two equal resistors, one of which is the labeled  $R$ .



open circuit

(zero current)



 (a) Long after the switch is closed

Inductor  $\approx$  short. In all three, the branch with the inductor (or the explicit wire in (1)) shorts the resistor branch  $\Rightarrow$  no voltage across  $R$ .

$$(1) = (2) = (3) \quad (I_R = 0)$$

 Partially

(b) Just after the switch is reopened (after being closed a long time)

The inductor current cannot change instantaneously; it must circulate through whatever resistive path remains:

- (1): Current circulates through the explicit short, not through  $R \Rightarrow I_R = 0$ .
- (2): All inductor current goes through the single  $R \Rightarrow$  largest  $I_R$ .
- (3): Inductor current splits equally between the two equal resistors  $\Rightarrow I_R$  is half of (2).

$$(2) > (3) > (1)$$

 (c) Long after it is reopened

Inductor current has decayed to zero  $\Rightarrow$  no current anywhere.

$$(1) = (2) = (3) \quad (I_R = 0)$$

Q9 b)

9 Figure 30-29 shows three circuits with identical batteries, inductors, and resistors. Rank the circuits, greatest first, according to the current through the resistor labeled  $R$  (a) long after the switch is closed, (b) just after the switch is reopened a long time later, and (c) long after it is reopened.

Fig. 30-28 Question 8.

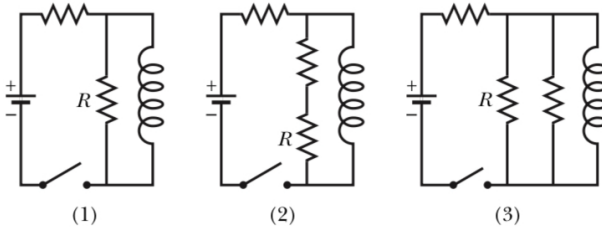
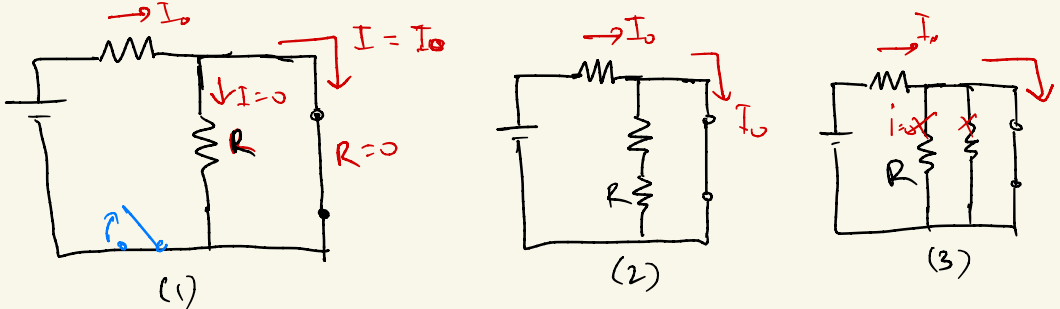


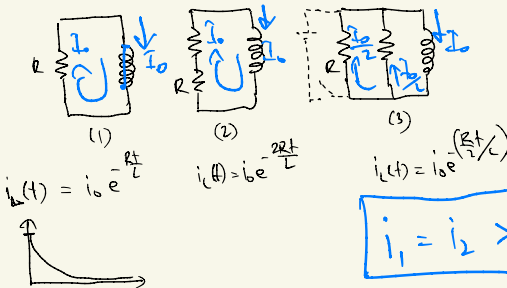
Fig. 30-29 Question 9.

a)



$$i_{R(1)} = i_{R(2)} = i_{R(3)} = 0$$

b)



$$i_1 = i_2 > i_3$$

•••59 SSM WWW In Fig. 30-64, after switch  $S$  is closed at time  $t = 0$ , the emf of the source is automatically adjusted to maintain a constant current  $i$  through  $S$ . (a) Find the current through the inductor as a function of time. (b) At what time is the current through the resistor equal to the current through the inductor?

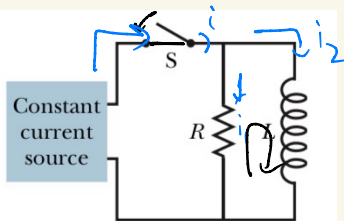


Fig. 30-64 Problem 59.

59. (a) We assume  $i$  is from left to right through the closed switch. We let  $i_1$  be the current in the resistor and take it to be downward. Let  $i_2$  be the current in the inductor, also assumed downward. The junction rule gives  $i = i_1 + i_2$  and the loop rule gives  $i_1 R - L(di_2/dt) = 0$ . According to the junction rule,  $(di_1/dt) = -(di_2/dt)$ . We substitute into the loop equation to obtain

$$L \frac{di_1}{dt} + i_1 R = 0.$$

This equation is similar to Eq. 30-46, and its solution is the function given as Eq. 30-47:

$$i_1 = i_0 e^{-Rt/L},$$

where  $i_0$  is the current through the resistor at  $t = 0$ , just after the switch is closed. Now just after the switch is closed, the inductor prevents the rapid build-up of current in its branch, so at that moment  $i_2 = 0$  and  $i_1 = i$ . Thus  $i_0 = i$ , so

$$i_1 = i e^{-Rt/L}, \quad i_2 = i - i_1 = i(1 - e^{-Rt/L}).$$

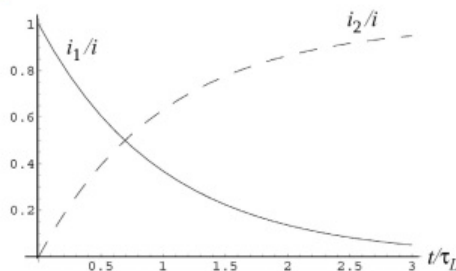
(b) When  $i_2 = i_1$ ,

$$e^{-Rt/L} = 1 - e^{-Rt/L} \Rightarrow e^{-Rt/L} = \frac{1}{2}.$$

Taking the natural logarithm of both sides (and using  $\ln(1/2) = -\ln 2$ ) we obtain

$$\left(\frac{Rt}{L}\right) = \ln 2 \Rightarrow t = \frac{L}{R} \ln 2.$$

A plot of  $i_1/i$  (solid line, for resistor) and  $i_2/i$  (dashed line, for inductor) as a function of  $t/\tau_L$  is shown below.

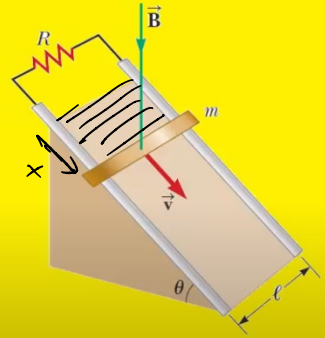


The figure below shows a bar of mass  $m = 0.200 \text{ kg}$  that can slide without friction on a pair of rails separated by a distance  $l = 1.20 \text{ m}$  and located on an inclined plane that makes an angle  $\theta = 25.0^\circ$  with respect to the ground. The resistance of the resistor is  $R = 1.00 \Omega$ , and a uniform magnetic field of magnitude  $B = 0.500 \text{ T}$  is directed downward, perpendicular to the ground, over the entire region through which the bar moves. With what constant speed  $v$  does the bar slide along the rails?

↑  
Equilibrium

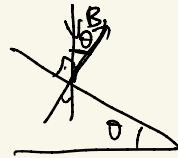
$$F_{\text{net}} = ma \rightarrow a = 0$$

$$\underline{F_{\text{net}} = 0}$$



$$\mathcal{E} = - \frac{d\Phi_B}{dt}$$

$$\begin{aligned}\Phi_B &= \vec{B} \cdot \vec{A} \\ &= BA \cos \theta \\ &= Blx \cos \theta\end{aligned}$$



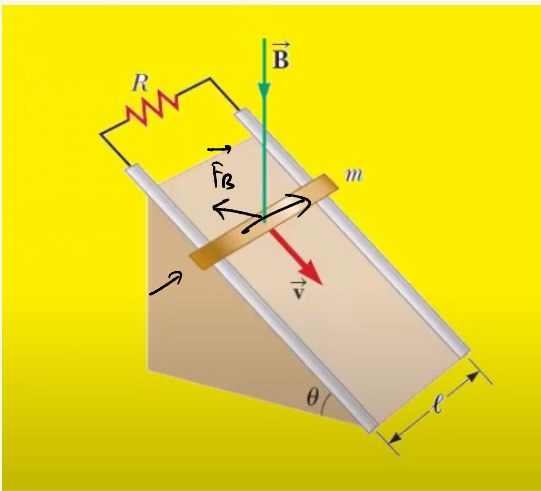
$$\mathcal{E} = - \frac{d\Phi_B}{dt} = - \frac{d}{dt} (Blx \cos \theta) = - Bl \cos \theta \frac{dx}{dt}$$

$$\mathcal{E} = - Blv \cos \theta$$

$$\boxed{|\mathcal{E}| = Blv \cos \theta}$$

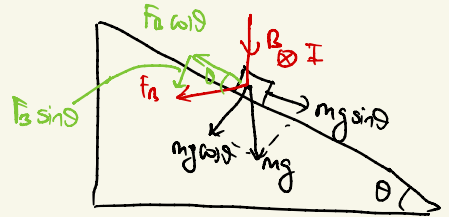
Current going through  $\mathcal{E} = IR \rightarrow I = \frac{\mathcal{E}}{R}$

$$\boxed{I = \frac{Blv \cos \theta}{R}}$$



$$\vec{F}_B = \vec{I} \vec{L} \times \vec{B}$$

$$F_B = \frac{Blv \cos \theta}{R} l B \sin \theta$$



For constant speed

$$F_B \cos \theta - mg \sin \theta = ma = 0$$

$$F_B \cos \theta = mg \sin \theta$$

$$\left( \frac{Blv \cos \theta}{R} l B \right) \cos \theta = mg \sin \theta$$

$$v = \frac{mgR \sin \theta}{B^2 l^2 \cos^2 \theta}$$

## HW Ch 30

Pr. 11, 23, 28, 29, 34, 42, 46, 47-48, 54, 57,  
75, 76, 77, 91, 95

Deadline: Sun Aug 24 noon 12:00 pm